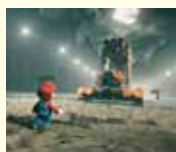


THROUGH THE LOOKING GLASS



While we all wait with bated breath for the retail entry of Google Glass, Verge's Joshua Topolsky visits Google's New York office and has a good hands on time with the Glass. Here is the report <http://dgit.in/13aOjdo>

SUPERMARIO IN 3D?



We have all played Super Mario and we love

it. In this fun video we look at Super Mario as an FPS in state-of-the-art 3D. It's a fun rendition of a classic. <http://dgit.in/W8ts8k>

TOO LITTLE, TOO LATE?

The Yahoo homepage redesign is no mean feat as it has stayed the same practically for a lot of years. In this article the author analyses how Marissa Mayer has done the unthinkable, but at the same time how it is just not enough. <http://dgit.in/XBtXTc>

HUMANS GOING EXTINCT?

This article discusses how mass extinctions have happened in the past and how the human race is constantly designing new ways to bring apocalypse with nuclear weapons, overdependence on computer networks, etc.. <http://dgit.in/126iNNO>

Tegra 4 CPU and GPU detailed: Nvidia is back in the smartphone saddle

- By Joel Hruska



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Tegra 4 has drawn eyeballs since Nvidia first unveiled the architecture at CES 2013, but concrete information has been scarce. Nvidia has broken its silence on the new chip and released technical whitepapers describing the CPU, GPU, and camera. We explore the processor and graphics options for the new microarchitecture, and the difference between the Tegra 4 (Cortex-A15) and Tegra 4i (Cortex-A9 R4).

TEGRA 4/4I'S GPU: THE EVOLUTION OF TEGRA 3

Last year, we predicted that Tegra 4 would be the first mobile GPU to take advantage of Nvidia's unified graphics architecture and potential CUDA support. At the time, it made sense — Kepler really moved the bar for performance per watt, and T4 seemed to be a likely candidate. That's not what Nvidia has done here. Instead, Tegra 4 is an updated and refined version of the GPU in Tegra 3, with some additional capabilities and performance under the hood. Tegra 2 was an eight-core design, split

between four pixel and four vertex units. Tegra 3 kept T2's single block of four vertex units, but doubled the pixel shader units (eight total). Tegra 4 adds five more vertex units (total of six) and two more pixel units (total of four) — but it also increases the number of cores per pixel unit.

The total count of 72 cores for a full T4 implementation breaks down to 48 pixel shaders and 24 vertex units. Other enhancements include full support for MSAA (previous Tegra chips only supported coverage sample antialiasing) and twice the number of raster operators (ROPs). Tegra 4 is capable

Tegra 4 vs Tegra 3 GPU stats

Tegra 4 / Tegra 3	
Vertex Shader	6x
Fragment ALU	8x
Pixel Rate	2.0x
Texture Rate	2.0x
Memory Rate	2.0x
3-kill Rate	1.3x

Tegra 4 - 72 Core GPU @ 872 MHz
4 pixel pipes * 3 ALU/pixel * 4 NAO/ALU = 6 VP/PA * 4 NAO/VP/PA

Tegra 3 - 32 Core GPU @ 800 MHz
2 pixel pipes * 3 ALU/pixel * 4 NAO/ALU = 1 VP/PA * 4 NAO/VP/PA

of outputting four pixels per clock, twice the rate of Tegra 3. NV claims that this more than doubles Tegra 3's fill rate. Tegra 4 also supports 4K x 4K textures, 16:1 Z compression and 4:1 color compression for optimum bandwidth conservation.

After reading through Nvidia's documentation, it's clear that Tegra 4's GPU is an evolutionary step past Tegra 3 rather than a fundamentally different processor. That's not necessarily a bad thing — T4's doubled ROPs and substantially increased vertex and pixel shaders should give the chip's performance a sizable kick in the pants. It also smooths the issue of backward compatibility with previous games and gives Nvidia time to work on a significant overhaul. I still think a programmable GPU architecture for Tegra makes sense in the long term, even if the company isn't ready to make that transition.

Tegra 4i's GPU is designed a bit differently. The integrated chip cuts half of T4's vertex shaders and doubles up the function units into two blocks instead of four. Memory bandwidth is also constrained to a single 32-bit channel rather than T4's 32-bit dual-channel design. Both of the new T4 chips support up to DDR3-2133. The memory controller itself has been beefed up in both Tegra 4/4i; the whitepaper notes that it includes

